Chapter

Cross-Domain Knowledge Transfer in Large Models: Transforming Knowledge Management Dynamics

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Abstract

In today's rapidly evolving technological landscape, large models—particularly large language models—are reshaping the dynamics of knowledge management. In this chapter, we explore how these advanced models facilitate cross-domain knowledge transfer—a critical capability in an era marked by increasingly interconnected global challenges and data overload—enabling the integration of information from diverse research fields. By leveraging the capabilities of large language models (LLMs), organizations can break down silos, foster innovation, and enhance decision-making processes. We systematically examine the mechanisms through which cross-domain knowledge transfer occurs in large language models, discuss the challenges such as data privacy and model bias, and highlight the implications for businesses operating in transformative environments. Case studies and current research provide insights into optimizing knowledge management strategies to harness the full potential of artificial intelligence in a multidisciplinary context. Overall, we hope that this chapter provides a foundational framework for scholars studying human-artificial intelligence (AI) collaboration and offers practitioners criteria to evaluate LLM adoption risks. We believe the findings in this chapter are critical for organizations seeking to balance agility with ethical accountability in the upcoming artificial general intelligence (AGI) era.

Keywords: knowledge management, cross-domain knowledge transfer, large language models, transformative environment, artificial intelligence, machine learning

1. Introduction

The fragmentation of knowledge across organizational silos and disciplines has become a key bottleneck for innovation in the twenty-first century [1]. As industries face pressures to adapt to rapid technological shifts and sustainability demands, traditional knowledge management (KM) systems struggle to bridge the gap between specialized expertise and cross-functional collaboration [2–4]. Knowledge management (KM) has traditionally relied on structured repositories,

workflows, and human expertise to organize and disseminate information. However, the rise of artificial intelligence has disrupted these practices, bringing about a paradigm shift in how organizations approach the creation, storage, and utilization of knowledge [5].

The rapid integration of generative artificial intelligence (GenAI), that is, large language models (LLMs), into organizational workflows has created unprecedented opportunities and challenges for KM [6–8]. While existing studies have documented GenAI's technical capabilities, few have systematically explored how GenAI fundamentally redefines KM paradigms across industries or evaluated its long-term implications for interdisciplinary collaboration and decision-making. This research addresses this gap by analyzing the transformative role of GenAI in KM, with a focus on LLMs, to provide actionable frameworks for scholars and practitioners navigating this evolving landscape [9, 10]. This study addresses the following research questions:

- How do LLMs overcome the limitations of traditional KM systems in integrating tacit and cross-domain knowledge?
- What ethical and operational risks emerge from LLM-driven KM, and how can they be mitigated?
- What frameworks can guide organizations in balancing the democratization of expertise with governance requirements?

By addressing these gaps, this research contributes to both academic discourse and practical KM strategies. For scholars, it offers a taxonomy of LLM capabilities in cross-domain contexts and identifies underexplored risks such as epistemic dependency on AI-generated content. For practitioners, the proposed frameworks empower organizations to navigate the tension between innovation and governance, ensuring LLMs drive sustainable value rather than operational fragility.

1.1 The historical evolution of knowledge management systems

Knowledge management (KM) as a formal discipline emerged in the late twentieth century, emphasizing the collection, storage, and dissemination of knowledge [4, 5]. Early systems primarily relied on databases and hierarchical structures, which were excellent at preserving explicit knowledge—facts, procedures, and policies that could be codified. However, these rigid systems often resulted in the development of organizational silos, where departments and teams operated in isolation, limiting cross-functional collaboration and knowledge exchange [8, 9].

Intranets, wikis, and collaborative tools that emerged in the early 2000s sought to mitigate these challenges by enabling more fluid sharing of information [10]. For instance, platforms like SharePoint allowed employees to access centralized repositories, while tools such as Confluence encouraged team-driven content creation. Despite these advancements, traditional KM systems still struggled to integrate tacit knowledge—the intuitive, experience-based insights often held by individuals—and bridge knowledge gaps across departments, industries, or disciplines. These limitations, compounded by the exponential growth of unstructured data in the 2010s, exposed critical vulnerabilities in conventional KM approaches and underscored the need for adaptive, AI-powered solutions [10–13].

1.2 The role of AI in augmenting traditional knowledge processes

Artificial intelligence (AI) has transformed KM by introducing advanced capabilities such as automation, pattern recognition, and real-time insights [2, 3, 6]. Unlike earlier systems, which required manual input and organization, AI tools powered by machine learning and natural language processing (NLP) can autonomously analyze unstructured data, extract key insights, and provide contextual recommendations.

For example, modern AI-driven KM systems can process vast amounts of data—from emails and reports to social media—and identify trends or risks in real-time [7–9]. In industries like healthcare, this capability has enabled organizations to predict patient outcomes by synthesizing clinical records with research data. Similarly, in the financial sector, AI tools are used to conduct comprehensive risk assessments by combining historical economic data with current geopolitical developments. Furthermore, AI's ability to adapt and learn ensures that KM systems remain relevant in dynamic environments. For instance, chatbots and virtual assistants can continuously improve their responses by learning from user interactions, offering a personalized knowledge experience for employees and customers alike. By pushing KM beyond static information storage, AI facilitates a transition toward dynamic, real-time learning environments [11, 12].

However, current research predominantly focuses on narrow applications of AI in KM (e.g., document classification or chatbot deployment), overlooking systemic questions about scalability, ethical governance, and the integration of human-AI hybrid intelligence. While prior work has focused on automating discrete KM tasks, it often assumes AI's role as a passive tool rather than an active driver of organizational transformation. This overlooks the systemic shifts required to integrate human and machine intelligence in dynamic environments [10, 12, 13]. This study seeks to expand this discourse by examining how AI reshapes KM holistically—not just as a tool but as a catalyst for reimagining organizational learning and innovation.

1.3 Introducing large language models (LLMs)

Large language models (LLMs) represent a groundbreaking advancement in AI-driven knowledge management [2, 3]. Unlike traditional systems that rely on predefined rules and structured datasets, LLMs leverage vast and diverse datasets encompassing text, images, and other modalities to understand context, generate insights, and synthesize knowledge across domains.

For example, LLMs like GPT-4, DeepSeek, and Llama can process technical documents, business reports, and scientific studies simultaneously, extracting commonalities and actionable insights that drive decision-making. This cross-domain synthesis is particularly valuable in multidisciplinary environments, where knowledge from multiple fields must be integrated to address complex challenges.

One real-world application of LLMs is in pharmaceutical research. Here, models can analyze clinical trial data, patient feedback, and molecular biology research to identify promising drug candidates. Similarly, in the field of education, LLMs are being used to create adaptive learning systems that tailor content to individual learning styles and progress levels.

What truly sets LLMs apart is their ability to generate new knowledge by identifying patterns and connections that may not be immediately apparent to human analysts. For instance, an LLM can combine insights from environmental science and economics to propose sustainable business strategies that balance profitability with

ecological impact. This capacity for fluid integration and adaptation makes LLMs indispensable tools for organizations looking to thrive in complex, rapidly evolving landscapes [12, 14].

Our analysis shows that LLMs make three key contributions to KM: (1) cross-domain knowledge accessibility, enabling seamless knowledge sharing across isolated fields; (2) context-aware knowledge integration, synthesizing real-time data and contextual signals for optimized decision-making; (3) governance and risk mitigation, addressing challenges like bias and privacy issues. We stress the need to balance AI efficiency with human oversight, and emphasize ethical governance and sustainable practices to fully realize the potential of LLMs in KM.

These contributions redefine KM paradigms: LLMs are not merely tools but active agents of organizational transformation. For researchers, this work establishes a framework to study human-AI symbiosis in cross-domain contexts; for practitioners, it provides actionable guidelines to balance innovation with ethical governance.

1.3.1 Structure of the chapter

This chapter explores the transformative role of large language models (LLMs) in knowledge management (KM) through three key contributions: cross-domain knowledge accessibility, context-aware knowledge integration, and governance and risk mitigation. Each section provides technical evaluations, real-world applications, and governance strategies.

- *Section 2*: Examines the architectural foundations of LLMs (e.g., transformer models and few-shot learning) that enable cross-domain knowledge retrieval and processing.

- *Section 3*: Explores how LLMs break down organizational silos through case studies in healthcare (e.g., integrating clinical data with patient feedback) and manufacturing (e.g., unifying engineering and marketing insights).

- *Section 4*: Confronts ethical and operational risks of LLMs, proposing solutions such as federated learning for data privacy, adversarial debiasing for fairness, and transparency frameworks for accountability.

- *Section 5*: Highlights the implications for business decision-making, such as financial institutions using LLMs to predict market risks by correlating geopolitical events with historical trends.

- *Section 6*: Outlines future directions, including personalized knowledge systems, green AI practices, and decentralized architectures, to ensure responsible scaling of LLM applications.

- *Section 7*: Critically synthesizes the findings in relation to prior literature, highlights new insights, and demonstrates how this study contributes to the broader discourse on AI-driven KM.

- *Section 8*: Concludes the study by summarizing the theoretical and managerial implications, outlining future research directions, and providing a call to action for ethical governance and sustainability in LLM-driven KM.

By addressing these themes, the chapter provides a comprehensive framework for scholars and practitioners to harness the transformative potential of LLMs while navigating their challenges.

2. Understanding large models and cross-domain knowledge transfer

2.1 Architecture and capabilities of large models

Large language models (LLMs), such as OpenAI's GPT series or Google's BERT, are sophisticated neural networks designed to process and generate human-like text [2, 3]. At their core lies the transformer architecture, which allows them to handle sequences of text with unparalleled efficiency. The transformer model uses attention mechanisms to identify relationships between words, phrases, or concepts within vast datasets, enabling it to understand and predict text contextually.

LLMs' strength lies in their scale. Trained on billions of data points across multiple domains, they develop a generalized understanding of language and knowledge that extends beyond isolated fields. These models can generate coherent essays, translate languages, summarize texts, and even create programming code, showcasing their versatility and cross-functional capabilities. An additional innovation is their ability to handle fine-tuning [7–9]. Organizations can train LLMs on specific datasets, tailoring their knowledge to meet specialized needs—from financial forecasting to molecular biology research—while retaining their broad foundational knowledge.

2.2 Mechanisms of cross-domain knowledge transfer

Cross-domain knowledge transfer occurs when insights or patterns identified in one field are adapted and applied to another [4–6]. For example, techniques from computational biology can inspire innovations in AI-driven diagnostics, or supply chain optimization methods can inform pandemic response strategies. LLMs are uniquely equipped for this task because they:

1. *Recognize shared patterns:* LLMs analyze diverse datasets, identifying recurring principles or analogies across fields. For instance, a model trained on economic theory and climate data might find parallels in resource scarcity models.

2. *Abstract core concepts:* By generalizing knowledge, LLMs distill complex ideas into universal principles that are applicable across disciplines. For example, concepts of risk assessment in finance can inform project management strategies in engineering.

3. *Facilitate communication:* LLMs bridge the gap between disciplines by translating technical jargon into universally understandable language. This makes them ideal for interdisciplinary teams working on complex problems.

One mechanism driving this capability is "few-shot learning," where LLMs adapt to new tasks with minimal additional data [13, 14]. For example, when exposed to a handful of examples from a niche domain, the model quickly grasps the context and applies its pre-existing knowledge to generate relevant outputs.

2.3 Real-world examples

1. *Pharmaceutical innovation:* Pharmaceutical companies leverage LLMs to analyze medical literature, clinical trial data, and patient feedback to identify potential treatments. For example, Pfizer's AI-driven drug discovery programs use models to cross-reference molecular biology findings with real-world patient outcomes, accelerating research timelines significantly.

2. *Business sustainability:* Corporations are using LLMs to combine insights from environmental science and business data to craft sustainable strategies. For instance, a major retailer integrated climate models and economic predictions to optimize their supply chain, reducing carbon emissions by 30.

3. *Multidisciplinary academic research*: Universities increasingly rely on LLMs to facilitate collaboration among departments. In one case, an AI research team worked with environmental scientists to create predictive models for wildfire spread by integrating climate data with historical wildfire patterns. This interdisciplinary approach enabled quicker response strategies and more accurate risk assessments.

4. *Financial risk management:* Financial institutions are deploying LLMs to merge economic indicators, geopolitical news, and historical market trends to provide real-time risk analysis. For example, JPMorgan's AI systems identify emerging risks by synthesizing seemingly unrelated data—such as international trade agreements and weather forecasts—to predict potential market disruptions.

5. *Expanding boundaries with LLMs:* As LLMs become more sophisticated, their ability to enable cross-domain knowledge transfer continues to grow. By integrating datasets from multiple disciplines, these models facilitate innovations that were previously unimaginable. In the sections ahead, we will explore how these capabilities address organizational challenges and drive advancements in sectors ranging from healthcare to education.

6. *Architecture and capabilities of large models:* Large models, such as GPT and BERT, are built using neural networks that mimic human reasoning. The transformer architecture, specifically, empowers these models to handle sequences of text, making connections between disparate pieces of information.

7. *Mechanisms of cross-domain knowledge transfer:* Cross-domain transfer happens through the model's ability to generalize patterns. For example, knowledge from biology can inform medical device design, while insights from environmental science can guide business sustainability efforts. LLMs learn to abstract core principles and apply them across varying contexts.

Overall, organizations use LLMs for tasks such as drafting multidisciplinary reports, predicting market behavior, and synthesizing cross-industry trends. A pharmaceutical company, for instance, could leverage LLMs to combine chemical research with healthcare data for innovative solutions.

3. Breaking down silos: Integrating knowledge across domains

3.1 Overcoming challenges of knowledge silos

Organizational silos, long seen as a barrier to efficiency and innovation, arise when departments or teams operate in isolation [7, 10]. This fragmentation often leads to duplication of efforts, gaps in communication, and a failure to leverage shared expertise. Historically, silos have been perpetuated by cultural, structural, and technological factors. For example, different divisions within a company may use incompatible software systems, making cross-departmental collaboration cumbersome [1]. LLMs are poised to dismantle these barriers by creating centralized hubs where all data and knowledge converge [2, 3].

Through natural language processing (NLP) capabilities, LLMs can ingest and synthesize information from disparate systems, generating cohesive insights that are accessible to everyone within an organization. For instance, a global manufacturing firm struggling with disjointed processes across its R&D, production, and marketing departments could use LLMs to unify knowledge. The model would harmonize data from technical manuals, supply chain records, and market research reports, enabling teams to work with a shared understanding.

In addition, LLMs foster transparency by making organizational knowledge universally searchable and retrievable. They act as a bridge between teams, ensuring that critical insights are not lost in isolated systems but instead contribute to the collective intelligence of the organization.

3.1.1 Facilitating interdisciplinary collaboration

True innovation often occurs at the intersection of disciplines. However, facilitating collaboration between teams with distinct expertise—such as engineers and marketers, or scientists and policymakers—can be challenging due to differences in terminology, methodologies, and priorities. LLMs excel in translating complex, technical jargon into accessible language, enabling diverse groups to collaborate effectively.

For example, in the healthcare industry, a team of medical researchers, software engineers, and data analysts might work together to develop an AI-driven diagnostic tool. Each group brings unique expertise, but their differing vocabularies can create misunderstandings. An LLM can mediate this collaboration by generating plain-language summaries of technical specifications, aligning goals, and even proposing actionable steps based on the collective input.

Another way LLMs facilitate interdisciplinary work is by providing contextual insights. A marketing team launching a new product might need input from the engineering department about manufacturing feasibility. Instead of wading through dense technical documents, the marketers can use an LLM to generate concise summaries that highlight relevant details, streamlining communication and decision-making.

Case study 1: Streamlining Product Development Cycles A multinational consumer electronics company faced significant delays in product development due to a lack of coordination between R&D, supply chain, and marketing teams. By implementing an LLM-driven knowledge management platform, the company created a centralized repository of insights that all departments could access [1]. For example, R&D teams could instantly retrieve market feedback analyzed by the marketing department,

while supply chain managers could align inventory forecasts with new product launches. This integration reduced product development cycles by 25%, saving both time and resources.

Case study 2: Advancing Climate Action through Collaboration A consortium of environmental organizations and government agencies sought to develop actionable strategies for mitigating climate change. The project required expertise from climatologists, economists, urban planners, and policymakers. Using an LLM, the consortium synthesized data from diverse sources, such as scientific research papers, economic models, and policy briefs. The model also generated multilingual summaries for stakeholders worldwide, enabling seamless collaboration across regions and disciplines. As a result, the consortium produced a comprehensive climate action plan in half the expected time.

Case study 3: Enhancing Supply Chain Resilience During the COVID-19 pandemic, a global logistics company experienced disruptions due to fluctuating demand and border closures. By deploying an LLM, the company integrated data from international trade agreements, weather forecasts, and real-time shipment tracking systems [1]. The model provided predictive insights into potential delays and recommended alternative routes, ensuring timely deliveries. This application not only improved operational efficiency but also strengthened relationships with clients by maintaining service reliability.

3.2 The transformative power of integration

LLMs are more than just tools for efficiency—they are enablers of cultural change within organizations. By breaking down silos, they promote a mindset of shared ownership and collaboration, where teams are empowered to work together toward common goals. For example, in a large consulting firm, the integration of an LLM-based platform transformed team dynamics, allowing consultants across global offices to share best practices and client insights in real-time, fostering a culture of innovation and shared expertise.

As businesses increasingly operate in complex, fast-changing environments, the ability to integrate knowledge across domains will become a defining factor for success. Companies that embrace LLMs for integration are more agile, adaptive, and competitive in responding to market demands.

3.3 Overcoming challenges of knowledge silos

Organizational silos restrict knowledge flow. Departments often work independently, limiting innovation [4, 10]. LLMs address this by creating centralized hubs where data from all departments converge. For instance, a financial institution adopted an LLM to unify fragmented customer data across departments, improving client engagement and significantly reducing service redundancies. Additionally, these models facilitate democratized access to knowledge. Employees at all levels can leverage LLMs to retrieve relevant insights, ensuring that decision-making is not limited to a few individuals with specialized knowledge.

3.4 Facilitating interdisciplinary collaboration

By translating complex technical jargon into accessible language, LLMs allow diverse teams—from engineers to marketers—to collaborate effectively. For instance,

in the energy sector, LLMs helped project managers align renewable energy engineers with policy analysts to design actionable plans for meeting sustainability goals. The LLM acted as a mediator, summarizing regulatory frameworks for engineers and technical blueprints for analysts, ensuring alignment across all stakeholders. They streamline workflows by acting as intermediaries, ensuring all parties work with synchronized information. Beyond this, LLMs can proactively highlight potential bottlenecks in collaborative projects by analyzing workflow data and suggesting solutions, enabling teams to address issues before they escalate.

Case study 1: Accelerating automotive innovation an automotive company faced delays in rolling out electric vehicle models due to disjointed communication between design and manufacturing teams. By deploying an LLM, the company created a shared knowledge platform that integrated feedback from both teams [1]. This reduced miscommunication, streamlined manufacturing processes, and accelerated time-to-market by 20%.

Case study 2: Advancing precision medicine a biotech firm used an LLM to integrate genetic research with clinical trial data, allowing researchers and healthcare professionals to collaborate more effectively. The platform's ability to synthesize these data sources helped identify promising therapies for rare diseases, significantly reducing the time needed for analysis.

Case study 3: Transforming retail logistics a global retailer implemented an LLM to coordinate logistics, merchandising, and customer support teams. By integrating real-time inventory data with customer feedback, the retailer improved stock management and reduced delivery delays by 30%. This holistic approach enhanced customer satisfaction and streamlined operations.

4. Challenges in cross-domain knowledge transfer

Data privacy and security: As organizations increasingly adopt large language models (LLMs) to enable cross-domain knowledge transfer, concerns around data privacy and security have grown significantly [11–13]. These models are often trained on extensive datasets that may include sensitive or proprietary information. For example, a healthcare organization utilizing an LLM to analyze patient records must ensure that the data is encrypted and anonymized to prevent potential breaches or unauthorized access. Beyond technical safeguards, organizations need robust policies to regulate how data is collected, stored, and shared.

One of the emerging strategies is the adoption of federated learning—a technique that allows models to be trained across decentralized datasets without transferring sensitive data to a central server. For instance, a consortium of hospitals could use federated learning to collaboratively improve diagnostic algorithms while preserving patient privacy. Furthermore, implementing zero-trust architectures and end-to-end encryption ensures that data remains secure throughout its lifecycle.

Addressing bias in models: Bias within LLMs poses a significant challenge, especially in cross-domain knowledge transfer, where diverse datasets with varying degrees of quality and representation are used. Models trained on biased or incomplete data risk perpetuating stereotypes or excluding critical perspectives, leading to flawed outputs. For example, an LLM used in recruitment might inadvertently favor candidates from overrepresented demographics due to historical biases in hiring data.

Addressing this issue requires a multipronged approach. First, datasets must be audited and curated to ensure diversity and fairness. Techniques like adversarial

debiasing and bias mitigation algorithms can be applied during the training process to reduce systemic distortions. Additionally, transparency in model decision-making is critical. Explainability tools, such as SHapley Additive exPlanations (SHAP) or Local Interpretable Model-agnostic Explanations (LIME), enable stakeholders to understand and scrutinize the reasoning behind an LLM's predictions, allowing biases to be identified and corrected.

Ethical and governance concerns: The rapid deployment of LLMs raises profound ethical questions. For instance, should an LLM-generated recommendation in a healthcare context carry the same weight as a decision made by a human professional? Similarly, how do we address the risks of misuse, such as generating misleading content or automating discriminatory practices?

To navigate these challenges, organizations must establish comprehensive AI governance frameworks. These should include ethical guidelines, accountability structures, and clear protocols for model deployment and usage. For example, a financial institution employing an LLM for credit risk assessment might require periodic audits to ensure compliance with anti-discrimination laws. Furthermore, organizations should engage diverse stakeholders—including ethicists, legal experts, and end-users—in the development and oversight of AI initiatives.

Emerging challenges and future directions: While current challenges like data privacy, bias, and ethics are well-documented, new issues are emerging as LLMs become more integrated into organizational workflows [6, 14]. One such challenge is the environmental impact of training and maintaining these models. The computational power required for LLMs contributes to significant carbon emissions, prompting organizations to explore greener AI practices. Techniques like model pruning, knowledge distillation, and the use of energy-efficient hardware are gaining traction as potential solutions.

Another frontier is the challenge of ensuring contextual accuracy in cross-domain applications. While LLMs are proficient at synthesizing information, they can sometimes generate outputs that are factually incorrect or contextually irrelevant. Reinforcing human oversight and incorporating domain-specific fine-tuning can mitigate these risks, ensuring that outputs align with organizational goals and standards.

By proactively addressing these challenges, organizations can unlock the full potential of LLMs while ensuring that their deployment aligns with ethical, legal, and societal expectations. The use of LLMs introduces concerns about data leaks and breaches. Sensitive organizational knowledge must be protected through robust encryption and ethical AI practices.

Addressing bias in large language models: Bias in training datasets can lead to flawed outcomes. For example, skewed historical data might reinforce existing inequalities. Organizations must actively audit and retrain models to mitigate biases. AI's growing role necessitates comprehensive governance frameworks. Misuse of LLMs, such as generating misleading content, must be countered through transparency and accountability.

5. Implications for business and decision-making

5.1 Enabling informed decision-making

Large language models (LLMs) are revolutionizing business decision-making by providing real-time, actionable insights derived from analyzing complex and

multidimensional datasets [2, 3]. Unlike traditional decision-support systems that rely on predefined algorithms and structured data, LLMs are capable of interpreting vast amounts of unstructured information, including text, audio, and even images. This capability allows organizations to uncover hidden patterns, predict trends, and make decisions with a level of confidence previously unattainable.

5.2 Enhanced predictive capabilities

One of the primary ways LLMs support informed decision-making is through enhanced predictive capabilities. By analyzing historical and real-time data, these models can forecast outcomes across multiple domains. For instance, in supply chain management, LLMs can predict potential disruptions by synthesizing data from weather forecasts, geopolitical developments, and market trends. Businesses can then proactively adjust their strategies, mitigating risks and minimizing losses.

In marketing, LLMs help organizations identify shifting consumer preferences by analyzing social media, customer feedback, and industry reports. These insights enable companies to design campaigns that resonate with their target audience, increasing engagement and return on investment.

5.3 Real-time contextual understanding

Unlike traditional systems, which often require human intervention to contextualize insights, LLMs can process and interpret data within their context. For example, an LLM integrated into a financial institution's operations can monitor global news, identify events that may impact the market (such as policy changes or natural disasters), and provide actionable recommendations tailored to specific investment portfolios. This capability ensures that decision-makers are always informed by the most relevant and timely information.

5.4 Democratizing decision-making

Another significant impact of LLMs is the democratization of decision-making. By making advanced analytical tools accessible to non-technical stakeholders, these models empower employees across all levels of an organization to contribute to strategic discussions. For instance, a project manager without a background in data science can use an LLM-powered dashboard to explore potential project risks and develop contingency plans based on data-driven insights.

5.5 Automating routine decisions

In addition to supporting high-level strategic decisions, LLMs are also transforming routine decision-making processes. For example, in e-commerce, LLMs can automate inventory restocking decisions by analyzing sales data, seasonal trends, and supplier availability. Similarly, in HR management, LLMs can assist with workforce planning by predicting staffing needs based on historical patterns and projected business growth.

As LLMs continue to evolve, their role in decision-making will expand, enabling businesses to navigate increasingly complex environments with greater agility and precision.

LLMs provide actionable insights by analyzing complex datasets in real-time. Businesses can make agile decisions, such as adapting supply chains to global events or tailoring marketing strategies to emerging trends.

5.6 Industry-specific applications

- Healthcare: LLMs are advancing precision medicine by integrating genomic data, clinical research, and patient feedback to identify personalized treatment options. For example, they are being used to predict patient outcomes and recommend therapies tailored to specific genetic profiles, significantly enhancing the efficacy of treatment plans.

- Finance: Financial institutions are leveraging LLMs for comprehensive risk assessment by synthesizing global economic indicators, policy changes, and historical market trends. They also aid in fraud detection by analyzing transaction patterns and identifying anomalies in real-time.

- Education: In education, LLMs are revolutionizing learning by creating adaptive learning paths that cater to individual student needs. By combining cognitive science with pedagogical data, they offer customized content delivery and real-time feedback to improve student engagement and outcomes.

- Manufacturing: LLMs are optimizing production processes by analyzing supply chain data, predicting equipment failures, and enhancing quality control through cross-referencing historical data with current performance metrics.

- Energy sector: Renewable energy companies use LLMs to integrate weather patterns, market trends, and policy updates to optimize energy production and distribution, ensuring efficiency and sustainability.

5.7 Driving innovation

LLMs are not only solving existing problems but also uncovering opportunities for innovation. By synthesizing vast amounts of interdisciplinary data, companies can identify emerging market trends and anticipate consumer needs. For example:

- Exploring new markets: Companies are using LLMs to analyze demographic, cultural, and economic data to identify opportunities in previously untapped regions.

- Developing cutting-edge products: Research and development teams use LLMs to generate design ideas, simulate product performance, and optimize prototypes based on customer feedback and market research.

- Adapting to disruptions: LLMs provide real-time analytics on external disruptions such as pandemics, natural disasters, or geopolitical shifts, allowing businesses to quickly pivot strategies and maintain operational continuity.

6. Future directions in AI and knowledge

Artificial intelligence and knowledge management are on the cusp of a transformative era, marked by several emerging trends that promise to redefine how organizations operate and innovate. These trends leverage the latest advancements in AI and LLMs to address complex challenges and unlock new opportunities [7, 12].

6.1 Personalized knowledge systems

As AI models become more sophisticated, the focus has shifted toward creating highly personalized knowledge systems. These systems use advanced machine learning algorithms to deliver tailored insights to individual users based on their specific roles, preferences, and real-time needs. For example, a sales manager could receive personalized market trend analyses, while a researcher might gain access to curated scientific papers relevant to ongoing projects. The integration of LLMs with user behavior analytics allows for dynamic adaptation, ensuring that recommendations remain contextually relevant and actionable.

6.2 Real-time learning and continuous updates

One of the most exciting advancements is the emergence of real-time learning systems. Traditional AI models are often static, requiring periodic updates to incorporate new data. In contrast, modern LLMs are increasingly capable of integrating continuous data streams, enabling them to learn and adapt in real-time. This capability is particularly valuable in industries such as finance and healthcare, where timely insights can significantly impact decision-making. For example, an LLM monitoring global financial markets can instantly adjust risk assessments in response to breaking news or policy changes.

6.3 Multimodal AI integration

The next generation of LLMs is designed to process and synthesize information across multiple modalities, including text, images, audio, and even video. This multimodal integration expands the potential applications of AI in knowledge management. For instance, in the medical field, an LLM could combine textual patient records with diagnostic imaging to provide a comprehensive analysis, enhancing diagnostic accuracy. Similarly, in manufacturing, models could analyze video footage of production lines alongside maintenance logs to predict equipment failures more effectively.

6.4 Decentralized AI architectures

As concerns around data privacy and computational efficiency grow, decentralized AI architectures are gaining traction. Federated learning, a technique where models are trained across distributed datasets without centralizing the data, is becoming a cornerstone of secure and efficient AI systems. Organizations are increasingly adopting this approach to ensure compliance with data protection regulations while still leveraging the power of collective intelligence. For example, a global network of hospitals could use federated learning to train a diagnostic AI model without compromising patient privacy.

6.5 AI-augmented collaboration platforms

AI-powered collaboration tools are revolutionizing how teams work together across geographies and disciplines. These platforms leverage LLMs to facilitate seamless communication, translate technical jargon, and provide real-time insights during collaborative projects. For instance, a cross-functional team developing a new product could use an AI tool to consolidate feedback from engineers, marketers, and designers, ensuring alignment and accelerating time-to-market.

6.6 Sustainability and green AI

The environmental impact of training and deploying LLMs has become a growing concern. The AI community is actively exploring strategies to reduce energy consumption, such as developing energy-efficient algorithms, utilizing renewable energy for data centers, and optimizing model architectures through techniques like pruning and distillation. Organizations are increasingly prioritizing these green AI practices to balance technological advancement with environmental responsibility.

6.7 Democratization of AI tools

The accessibility of AI tools is expanding, enabling even small and medium-sized enterprises (SMEs) to benefit from advanced technologies. Cloud-based LLM platforms and open-source models are reducing the barriers to entry, allowing businesses of all sizes to integrate AI into their knowledge management processes [7]. This democratization fosters innovation across diverse sectors, from local startups to nonprofit organizations addressing global challenges.

By embracing these trends, organizations can position themselves at the forefront of the AI revolution, leveraging cutting-edge technologies to drive innovation, improve decision-making, and achieve sustainable growth [6, 14].

- Personalized knowledge systems: Tailored insights for individual users, improving relevance and efficiency.
- Real-time learning: Continuous updates to models ensure adaptability to changing data landscapes.
- Multimodal AI: Combining text, images, and audio to enhance cross-domain applications.

Speculating on the future: In 10 years, knowledge management may become entirely AI-driven, with systems autonomously curating and delivering insights to decision-makers, drastically reducing human effort in repetitive tasks.

7. Discussion on LLMs with KM

This study has explored the transformative role of large language models (LLMs) in enabling cross-domain knowledge transfer and reshaping knowledge management (KM) dynamics. By addressing the research questions outlined in the introduction, this work advances the understanding of how LLMs overcome the limitations of

traditional KM systems, mitigate ethical and operational risks, and provide frameworks for balancing innovation with governance. Below, we critically synthesize our findings in relation to prior literature, highlight new insights, and demonstrate how this study contributes to the broader discourse on AI-driven KM.

7.1 Critical synthesis of findings and literature

Cross-domain knowledge transfer and LLMs: Prior research has extensively documented the technical capabilities of LLMs, such as natural language processing (NLP) and pattern recognition [2, 3]. However, existing studies have largely focused on narrow applications, such as document classification or chatbot deployment, without systematically examining how LLMs enable cross-domain knowledge transfer. This study bridges this gap by demonstrating how LLMs integrate tacit and explicit knowledge across traditionally isolated domains, such as healthcare, finance, and logistics. For instance, our case studies reveal that LLMs can unify clinical trial data with patient feedback in pharmaceutical research, or correlate geopolitical events with market trends in financial risk management. These findings align with the literature on knowledge integration but extend it by highlighting LLMs' unique ability to dynamically synthesize multidisciplinary data in real-time.

Overcoming knowledge silos: The fragmentation of knowledge across organizational silos has long been identified as a barrier to innovation [4]. Traditional KM systems, such as intranets and wikis, have attempted to address this challenge but often fail to integrate tacit knowledge or bridge domain-specific gaps. Our findings show that LLMs act as centralized hubs, harmonizing data from disparate sources (e.g., technical manuals and market research reports) and making organizational knowledge universally accessible. This capability aligns with the literature on collaborative KM [5] but introduces a novel dimension: LLMs not only facilitate knowledge sharing but also generate new insights by identifying patterns and connections that are not immediately apparent to human analysts.

Ethical and operational risks: While the literature has highlighted risks such as algorithmic bias [6] and data privacy breaches, this study provides actionable frameworks for mitigating these challenges. For example, we propose federated learning for data anonymization in healthcare and adversarial debiasing techniques to ensure fairness in recruitment algorithms. These solutions build on prior work but emphasize the need for proactive governance and stakeholder engagement, which have been underexplored in the context of LLM-driven KM.

7.2 New insights and contributions

This study advances the reader's understanding of the research problem in several key ways:

- Context-aware knowledge integration: Unlike traditional KM systems, which rely on static repositories, LLMs enable dynamic, real-time synthesis of knowledge. This capability is particularly valuable in multidisciplinary environments, where insights from multiple fields must be integrated to address complex challenges.

- Epistemic dependency: We introduce the concept of epistemic dependency—organizations' reliance on AI-generated content—as a critical theoretical construct. This concept challenges the traditional view of KM as a human-centric

process and calls for new frameworks to balance AI-driven efficiency with human oversight.

- Scalability and generalization: Our findings demonstrate that LLMs can be fine-tuned for domain-specific applications while retaining their cross-domain capabilities. This insight addresses a key limitation in the literature, which often assumes a trade-off between specialization and generalization.

7.3 Limitations and future research

While this study provides a comprehensive framework for understanding LLM-driven KM [6, 10–13], several limitations warrant further exploration:

- Epistemic trust: Future research should investigate how organizations can balance reliance on AI-generated insights with human critical thinking, particularly in high-stakes domains like healthcare and finance.

- Scalability vs. ethics: The trade-offs between model size, computational efficiency, and ethical considerations remain underexplored. Future studies should examine how these trade-offs impact equitable access to AI capabilities.

- Human-AI creativity: The role of LLMs in augmenting—rather than replacing—human creativity and innovation requires further investigation.

This study has demonstrated that LLMs are not merely tools for automating KM tasks but active agents of organizational transformation [5, 7–9]. By enabling cross-domain knowledge transfer, breaking down silos, and fostering interdisciplinary collaboration, LLMs redefine how organizations create, store, and utilize knowledge. However, realizing this potential requires addressing ethical, operational, and environmental challenges through proactive governance and stakeholder engagement. As organizations navigate the complexities of the AI era, this study provides a roadmap for harnessing the transformative power of LLMs while ensuring alignment with societal values and sustainability goals.

8. Conclusion and call to action

The transformative potential of large language models (LLMs) in enabling cross-domain knowledge transfer, breaking down silos, and fostering innovation across industries has been a central theme of this book [2, 3]. LLMs have emerged as powerful tools that redefine how organizations manage, integrate, and apply knowledge, offering unprecedented efficiency, adaptability, and scalability. Through real-world applications, we have demonstrated how LLMs streamline operations, facilitate interdisciplinary collaboration, and deliver actionable insights across diverse sectors such as healthcare, finance, education, and logistics. However, the journey toward leveraging LLMs effectively is not without challenges. Issues of data privacy, model bias, ethical concerns, and environmental impact remain critical areas that require proactive strategies. To fully harness the potential of LLMs, organizations must adopt a balanced approach that combines technological innovation with ethical governance and sustainability [6–8].

For simplicity, we structure it into three subsections to address the theoretical and managerial implications of the study, as well as to outline future research directions.

8.1 Theoretical implications

This research contributes to the theoretical understanding of knowledge management (KM) in the era of artificial intelligence by highlighting the role of LLMs as active agents of organizational transformation rather than passive tools. The study bridges a critical gap in the literature by systematically exploring how LLMs enable cross-domain knowledge transfer, a capability that has been underexplored in traditional KM frameworks. By introducing the concept of context-aware knowledge integration, this work provides a new lens through which to examine how LLMs dynamically synthesize information from diverse domains, enabling organizations to address complex, multidisciplinary challenges [12–14].

Furthermore, the study advances the discourse on human-AI collaboration by proposing a taxonomy of LLM capabilities in cross-domain contexts [7, 11]. It underscores the importance of epistemic dependency—the reliance on AI-generated content—as a theoretical construct that warrants further investigation. This research also highlights the need for interdisciplinary approaches to studying AI-driven KM, emphasizing the integration of insights from computer science, organizational behavior, and ethics to develop holistic frameworks for understanding and optimizing human-AI symbiosis.

For practitioners, this research offers actionable insights into how organizations can leverage LLMs to enhance decision-making, foster innovation, and maintain competitive advantage in transformative environments. Key managerial implications include:

1. *Breaking down silos:* Deploy LLMs as centralized hubs to integrate siloed data (e.g., R&D blueprints, customer feedback, and regulatory documents), enabling cross-functional teams to collaborate on shared insights.

2. *Ethical AI governance:* Implement auditable AI frameworks with bias mitigation tools (e.g., adversarial debiasing) and transparency protocols (e.g., SHAP-based explainability) to ensure compliance with regulations like GDPR and build stakeholder trust.

3. *Sustainability in AI deployment:* Prioritize energy-efficient practices (e.g., model pruning and federated learning) and renewable energy adoption to reduce LLMs' carbon footprint while maintaining performance.

8.2 Future research and call to action

8.2.1 Ideas for future research

To advance LLM-driven knowledge management, future studies should focus on:

- *Epistemic trust:* How can organizations balance reliance on AI-generated insights with human critical thinking? What frameworks ensure accountability in high-stakes decisions (e.g., medical diagnostics)?

- *Scalability vs. ethics:* How do trade-offs between model size (e.g., GPT-4's 1.7T parameters) and computational costs impact equitable access to AI capabilities?
- *Human-AI creativity:* Can LLMs augment—rather than replace—human innovation in fields like drug discovery or climate modeling?
- *Green AI:* Develop decentralized architectures (e.g., blockchain-integrated LLMs) to reduce energy consumption while preserving data privacy.
- *Societal equity:* Investigate policies to prevent AI-driven disparities in education, healthcare, and employment, ensuring marginalized communities benefit from LLM advancements.

The potential of LLMs to bridge disciplinary divides and address global challenges—from pandemic response to climate resilience—is unparalleled. Yet, this promise hinges on responsible innovation. Organizations must adopt proactive governance (e.g., ethics boards and impact assessments), invest in human-AI hybrid workflows, and prioritize inclusive design to ensure LLMs empower rather than marginalize. By uniting scholars, practitioners, and policymakers, we can steer AI toward a future where technology amplifies human potential and fosters collective progress.

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Abbreviations

LLM	large language model
KM	knowledge management
AI	artificial intelligence
AGI	artificial general intelligence
GenAI	generative artificial intelligence
NLP	natural language processing
SHAP	SHapley Additive exPlanations
LIME	Local Interpretable Model-agnostic Explanations
GDPR	General Data Protection Regulation
SME	small and medium-sized enterprises
GPT	generative pre-trained transformer
BERT	Bidirectional Encoder Representations from Transformers

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